

Natural Language Processing

Getting started with python, NLP libraries

2025/26

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1. Consider which computer, operating system and development environment you are going to use for this module. You will need a Python 3 installation, at least version 3.12
 - (a) Type 'python' or 'python3' in a terminal to check if you have python installed.
 - (b) If not found, download and install it from <https://www.python.org/downloads/>
 - (c) Type 'python --version' or 'python3 --version' to check the version number.
 2. The demonstrations in class will use the VSCode IDE. You can use any development environment you like, but we will only be able to give support for VSCode
 - (a) Check that you can open VSCode and run a simple Python program.
 - (b) Python and VSCode <https://code.visualstudio.com/docs/languages/python>
 - (c) Full guide to Python on windows <https://learn.microsoft.com/en-us/windows/python/beginners>
 - (d) Full guide to Python on MacOS <https://docs.python.org/3/using/mac.html>
 - (e) Full guide to Python on Linux <https://docs.python.org/3/using/unix.html>
 3. You will need to know the basics of interacting with your file system through a command prompt
 - (a) Brief guide to Windows Command Prompt <https://www.cs.princeton.edu/courses/archive/spr05/cos126/cmd-prompt.html>
 - (b) Unix/linux terminal commands <https://portal.cac.cornell.edu/techdocs/clusterinfo/linuxtips/>
 - (c) Command line environment for python <https://docs.python.org/3/using/cmdline.html>

4. NLTK Installation

- (a) Check if NLTK is installed

```
import nltk
sentence = "At eight o'clock on Thursday morning \
Arthur didn't feel very good."
tokens = nltk.word_tokenize(sentence)
print(tokens)
```

- (b) If nltk is not installed, follow the installation instructions for your platform:

<https://www.nltk.org/install.html>

- (c) Try some of the tokenization examples here: <https://www.nltk.org/howto/tokenize.html>

5. SpaCy installation

- (a) Check if spaCy is installed

```
from spacy.tokenizer import Tokenizer
from spacy.lang.en import English
nlp = English()
# Create a blank Tokenizer with just the English vocab
tokenizer = Tokenizer(nlp.vocab)

# Construction 2
from spacy.lang.en import English
nlp = English()
# Create a Tokenizer with the default settings for English
# including punctuation rules and exceptions
tokenizer = nlp.tokenizer
```

- (b) If spaCy is not installed, follow the installation instructions for your platform:

<https://spacy.io/usage>

- (c) Read the first six sections of this tutorial (up to and including 'vocab') and try the example code locally: <https://spacy.io/usage/spacy-101>

6. nltk and spaCy also require downloaded data for some functionality
You may run into permission problems here – ask for help if you can't find a solution

(a) NLTK data downloads

```
import nltk
nltk.download('popular')
```

(b) spaCy model download

Note: *this is a terminal command, not a python command*

```
python -m spacy download en_core_web_sm
```

...

... and back in python again

```
import spacy
nlp = spacy.load("en_core_web_sm")
doc = nlp("This is a sentence.")
print([(w.text, w.pos_) for w in doc])
```